

Power-of-Two Cycles in 24-Vertex Graphs with Degree Sequence $(5, 3^{23})$

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Abstract

We prove that every finite simple undirected graph with degree sequence $(5, 3^{23})$ contains a simple cycle of length 4, 8, or 16. The proof is an exact finite enumeration. Deleting the unique vertex of degree 5 and its neighbourhood reduces a putative counterexample to one of three classes of 18-vertex graphs. Nonisomorphic residual graphs are generated exhaustively, all attachment orbits are tested, and the graph parser, cycle detector, attachment enumeration, and three complete search streams are separately checked. The broader assertion for all 24-vertex graphs of minimum degree at least 3 remains unresolved: in particular, the argument does not cover degree sequence $(4, 4, 3^{22})$ or any larger degree excess.

1 Scope and relation to the finite boundary

For a graph G , let $\mathcal{C}(G)$ be its set of simple-cycle lengths. The Erdős–Gyárfás conjecture asks whether every graph of minimum degree at least 3 has a cycle whose length is a power of two. Garcia’s certified finite search proves that every such graph on at most 23 vertices has a 4- or 8-cycle [1, Theorem 2.1]. Garcia also records the 24-vertex cubic boundary arising from Markström’s exhaustive work [2, 1]. Thus order 24 is the next general finite layer, but its noncubic part is not settled by those results.

The degree excess over cubicity is

$$\text{exc}(G) = \sum_{x \in V(G)} (\deg_G(x) - 3).$$

At excess two there are two degree sequences, $(5, 3^{23})$ and $(4, 4, 3^{22})$. This note treats exactly the first. No claim of priority beyond this bounded comparison is made, and no assertion about the full 24-vertex minimum-degree-three problem is implied.

Theorem 1. *Every finite simple undirected graph on exactly 24 vertices in which one vertex has degree 5 and each of the other 23 vertices has degree 3 contains a simple cycle whose length belongs to $\{4, 8, 16\}$.*

2 A three-case reduction

We first isolate the mathematical reduction used by the enumeration.

Lemma 2. *Suppose that a graph G with degree sequence $(5, 3^{23})$ has no 4-cycle, and let v be its vertex of degree 5. Then $G[N(v)]$ is a matching of some size $m \in \{0, 1, 2\}$. After deleting $\{v\} \cup N(v)$, the residual graph H has 18 vertices, exactly $10 - 2m$ vertices of degree 2, exactly $8 + 2m$ vertices of degree 3, and $22 + m$ edges.*

Conversely, every graph G in the stated degree class and matching case m is obtained by attaching the degree-2 vertices of such an H to a five-vertex matching gadget as described below.

Proof. If one vertex of $N(v)$ had two neighbours in $N(v)$, those three neighbours together with v would form a 4-cycle. Hence $G[N(v)]$ has maximum degree at most one and is a matching of size $m \in \{0, 1, 2\}$.

No vertex outside $\{v\} \cup N(v)$ can be adjacent to two members of $N(v)$: those two incidences, together with their edges to v , would again form a 4-cycle. Each endpoint of the matching inside $N(v)$ therefore has one edge to H , whereas each unmatched member of $N(v)$ has two. The resulting $10 - 2m$ endpoints in H are distinct. Each loses one incident edge when the six vertices are deleted and hence has degree 2 in H ; every other vertex of H retains degree 3. Consequently

$$|E(H)| = \frac{2(10 - 2m) + 3(8 + 2m)}{2} = 22 + m.$$

For the converse, start with an 18-vertex graph having minimum degree 2, maximum degree 3, and $22 + m$ edges. If n_i is its number of degree- i vertices, then $n_2 + n_3 = 18$ and $2n_2 + 3n_3 = 44 + 2m$, so $n_2 = 10 - 2m$ and $n_3 = 8 + 2m$. Add a new vertex v adjacent to five new arm vertices, put a matching of size m on the arms, and attach every degree-2 vertex of H to exactly one arm. Give each matched arm one such attachment and each unmatched arm two. The resulting graph is simple and has degree sequence $(5, 3^{23})$. Every original graph in matching case m is recovered by taking its own residual graph and its induced assignment of ports to arms. $\square$

Call the degree-2 vertices of H its *ports*. Relabelling only the five-arm gadget reduces the assignments to

$$\frac{10!}{(2!)^5 5!} = 945, \quad \binom{8}{2} \frac{6!}{(2!)^3 3!} = 420, \quad \binom{6}{2} \frac{4!}{(2!)^2 2!} = 45$$

representatives for $m = 0, 1, 2$, respectively. These quotients cannot omit a graph: a gadget relabelling extends to an isomorphism fixing H and v . Automorphisms of H are deliberately not quotiented, so they can cause only repetition.

3 Exact enumeration

For each $m \in \{0, 1, 2\}$, the program **geng** from nauty 2.9.3 [3] generated one representative of every isomorphism class of simple 18-vertex graph with minimum degree 2, maximum degree 3, exactly $22 + m$ edges, and no 4-cycle. No connectedness option was imposed. A residual graph already containing an 8- or 16-cycle was discarded, since that cycle would persist after attachment. For every remaining residual graph, all attachment representatives above were formed and searched for cycles of lengths 4, 8, and 16, in that order.

The complete counts are shown in Table 1. The three rejection columns are sequential: an attachment counted as a C_4 rejection was not subsequently tested for C_8 or C_{16} . In every row their sum equals the attachment count, so no attachment survived.

Completion of the proof of Theorem 1. Assume that a counterexample G exists. It has no 4-cycle, so Lemma 2 places it in exactly one of the three matching cases. Its residual graph H occurs in the

Table 1: Exhaustive residual and attachment counts.

m	residuals	survivors	attachments	C_4	C_8	C_{16}
0	294,693	29,143	27,540,135	15,062,760	12,477,375	0
1	835,745	13,064	5,486,880	2,331,130	3,155,750	0
2	1,087,732	1,020	45,900	12,131	33,769	0

corresponding nonisomorphic **geng** stream. Because G has no target cycle, H passes the residual C_8 and C_{16} tests. Its port attachment is equivalent under a gadget automorphism to one of the tested representatives. The enumeration shows that every such representative has a target cycle, a contradiction. $\square$

4 Independent checks and reproducibility

The search source reads graph6 records, enumerates the attachment orbits, and detects an exact cycle length by depth-first enumeration of simple paths. The production detector uses a least-vertex convention and removes the two orientations of each cycle. The following checks were implemented separately from the relevant production component.

- Exhaustive labelled port permutations—all $10!$, $8!$, and $6!$ orders in the three cases—were canonicalised under the gadget automorphism groups. They gave exactly 945, 420, and 45 orbits and agreed with the production lists without a missing or duplicated representative.
- An unpruned reference cycle search agreed with the production detector in 57,234 comparisons. These included all 32,768 labelled graphs on six vertices for the C_4 test, cycle boundary fixtures of orders 3 through 24, and a fixed-seed collection of sparse and medium-density graphs.
- The production graph6 parser’s adjacency matrices agreed byte for byte with nauty’s **showg -a**. The initial deterministic samples and a separate full 35,063-graph shard both passed.
- A clean rebuild followed by a separate complete run of all three streams reproduced every count in Table 1, returned the documented exhaustion status, and produced no witness file.

The principal reproduction commands are:

```
c++ -std=c++17 -O3 -Wall -Wextra -pedantic \
geng_excess2_search.cpp -o geng_excess2_search

geng -q -f -d2 -D3 18 22:22 | \
./geng_excess2_search 0 pass2_A_m0.log pass2_A_m0.witness
geng -q -f -d2 -D3 18 23:23 | \
./geng_excess2_search 1 pass2_A_m1.log pass2_A_m1.witness
geng -q -f -d2 -D3 18 24:24 | \
./geng_excess2_search 2 pass2_A_m2.log pass2_A_m2.witness
```

The verifier returns status 20 on exhaustion and status 10 if it finds a witness. The four C++ sources and the three exhaustion logs accompany this article as computational supplementary material.

For comparison of proof architecture only, Zeng separates an analytic tail from exact, independently reproduced finite checks in a cycle-length probability problem [4]. The present theorem is wholly finite and uses none of that paper’s probabilistic claims.

5 Limitations and assistance disclosure

This is an exact computer-assisted result for one degree sequence, not an UNSAT certificate for the original 24-vertex problem. It leaves $(4, 4, 3^{22})$, every larger degree excess, and the unrestricted minimum-degree-three assertion untouched. The literature comparison was limited to the cited sources; it is not evidence of global priority.

An OpenAI Codex system assisted with the reduction, implementation and audit workflow, literature checking, and preparation of this manuscript. A separate audit rebuilt the programs, reran the exhaustive streams, checked the finite reduction, and cross-checked the components; such a model-based audit is not a substitute for peer review. The mathematical claim rests on the explicit reduction, source code, and exact runs described above, not on AI confidence or prose generation.

References

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